



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of Methodologies

- Data Collection via SpaceX API and web scraping
- Data Wrangling
- EDA with Data Visualization
- EDA with SQL
- Build and Interactive Map with Folium
- Build a Dashboard with Plotly Dash
- Predictive Analysis (Classification)

Summary of Results

- Launch success is influenced by factors such as launch site, orbit type, and payload mass, which are also interrelated.
- First-stage landing success rate has increased over time with more flights and newer booster versions.
- With machine learning models it is possible to predict if launch would be successful or not thus deciding cost of the launch.

Introduction

- **Project background and context**

Today, many companies are working to make space travel more affordable and accessible. Most aerospace companies charge around \$165 million for a rocket launch. However, SpaceX offers a Falcon 9 launch for about \$62 million.

The reason SpaceX can offer a much lower price is that it reuses the first stage of the Falcon 9 rocket. Since the first stage is the most expensive part of the rocket, recovering and reusing it helps reduce the cost of future launches.

If we can determine if the first stage will land, we can determine the cost of a launch. This information can be used if an alternate company wants to bid against SpaceX for a rocket launch.

- **Problems you want to find answers**
- Can we predict if the Falcon 9 first stage will land successfully?
- What factors affect the successful landing of the Falcon 9 first stage?

Section 1

Methodology

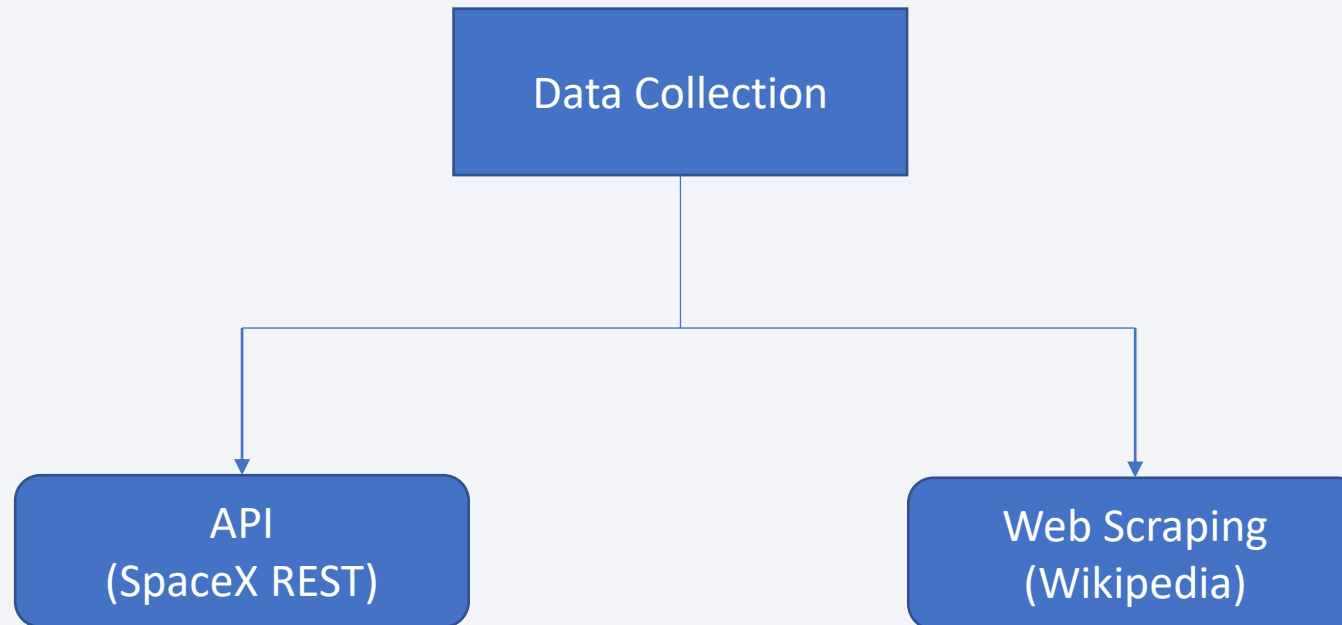
Methodology

Executive Summary

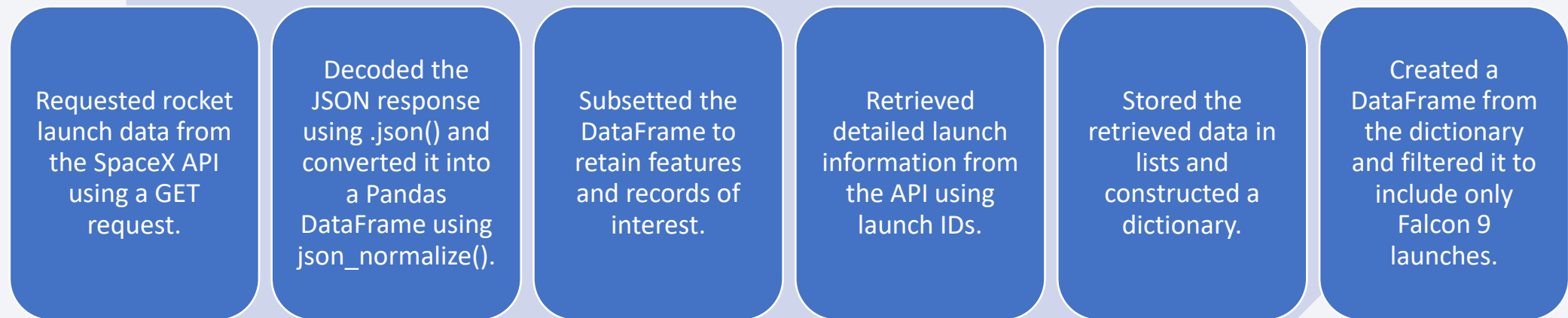
- Data collection methodology:
 - Falcon9 launch records were collected through API (SpaceX REST) and Web Scraping (Wikipedia).
- Perform data wrangling
 - Handled missing values using mean imputation and created the binary target variable (Class): 1 = successful landing, 0 = unsuccessful landing of Falcon9 first stage.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Performed predictive analysis using classification models by defining target and feature variables, standardizing data, developing and tuning models, evaluating performance, and selecting the best-performing model.

Data Collection

Data was collected from two sources: SpaceX REST API and Wikipedia web scraping.

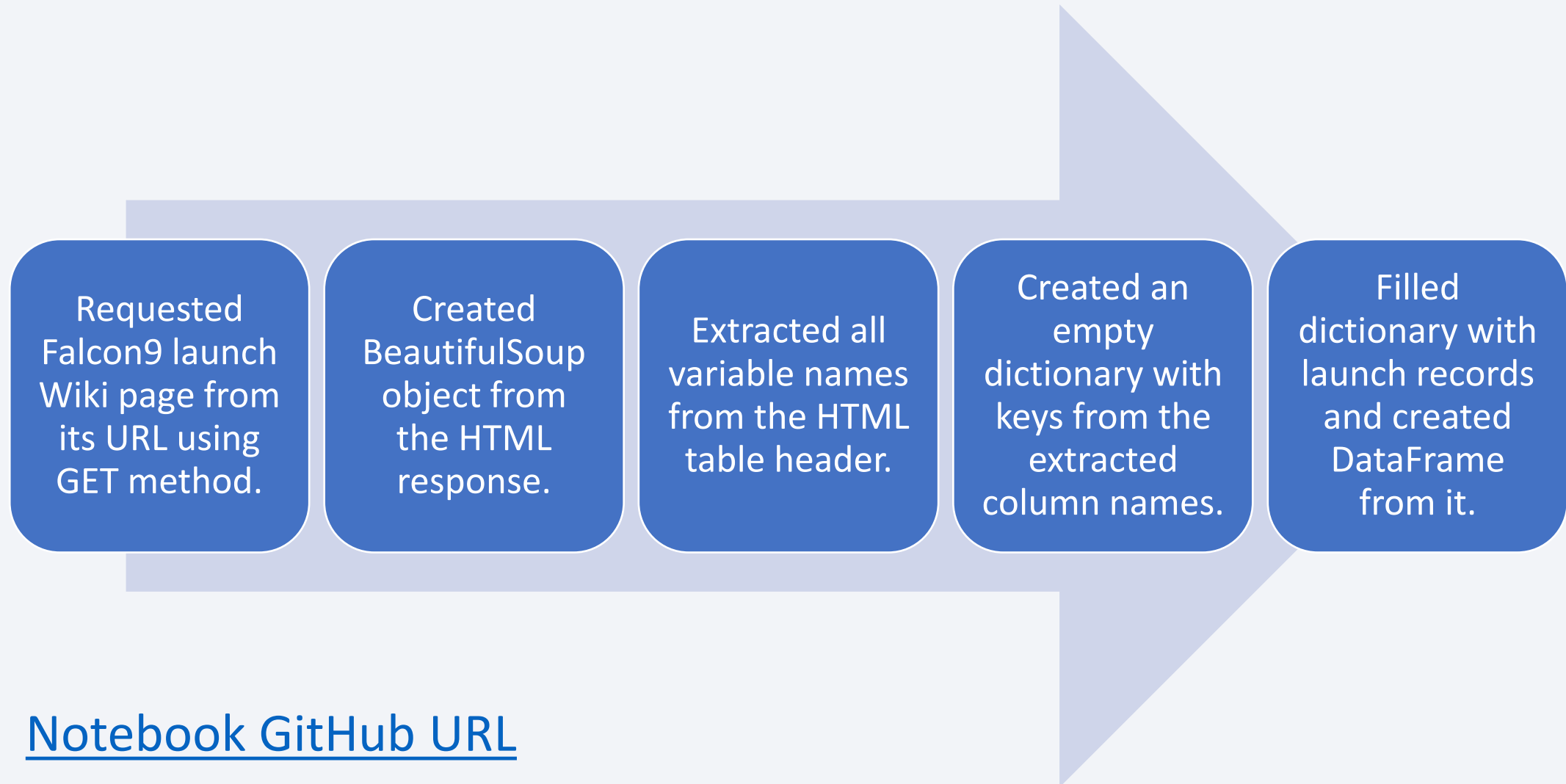


Data Collection – SpaceX API

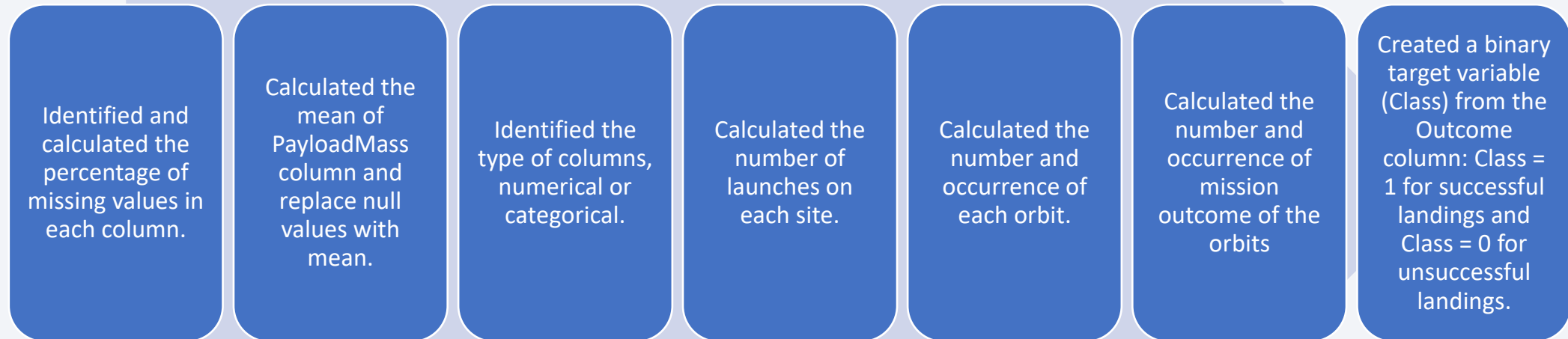


[Notebook GitHub URL](#)

Data Collection - Scraping



Data Wrangling



[Notebook GitHub URL](#)

EDA with Data Visualization

- A catplot of Flight Number versus Payload Mass was used to understand the effect of flight number and payload mass on the launch outcome.
- A catplot was used to compare the number of launches at each launch site and show whether the launches were successful or unsuccessful.
- A catplot of Payload Mass versus Launch Site was used to understand how payload mass differs across launch sites and its effect on the launch outcome.
- A bar chart was used to show and compare the launch success rate for each orbit type.
- A catplot was used to understand the relationship between Flight Number and Orbit Type and see their effect on the launch outcome.
- A catplot of Payload Mass versus Orbit Type was used to understand how payload mass differs across different orbit types and its effect on the launch outcome.
- A line chart of yearly launch success rate was used to understand the trend of launch success over time and identify changes in success rate across different years.

EDA with SQL

- Displayed the names of the unique launch sites in the space mission.
- Displayed 5 records where launch sites begin with the string 'CCA'.
- Displayed the total payload mass carried by boosters launched by NASA (CRS).
- Displayed average payload mass carried by booster version F9 v1.1.
- Listed the date when the first successful landing outcome in ground pad was achieved.
- Listed the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000.
- Listed the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.
- Ranked the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.
- Listed the total number of successful and failure mission outcomes.
- Listed all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

Build an Interactive Map with Folium

- Created and added `folium.Circle` and `folium.Marker` for each launch site to mark site locations on map.
- Added `folium.Marker` to a `MarkerCluster` for each launch result to group nearby markers and improve map readability.
- Added `MousePosition` to display the latitude and longitude of the mouse pointer for easier identification of map coordinates.
- Located the closest coastline point using `MousePosition` and calculated the distance between the launch site and the coastline and marked on map using `folium.Marker`.
- Drew a `PolyLine` between each launch site and coastline point to visually represent the calculated distance.

Build a Dashboard with Plotly Dash

- **Launch Site Dropdown Input Component**

- Added a dropdown menu to allow users to select a specific launch site and interactively explore launch performance.

- **Success Pie Chart Callback**

- Developed a callback function to dynamically update the success/failure pie chart based on the selected launch site.
- This helps compare the success rate of different launch sites.

- **Payload Range Slider Component**

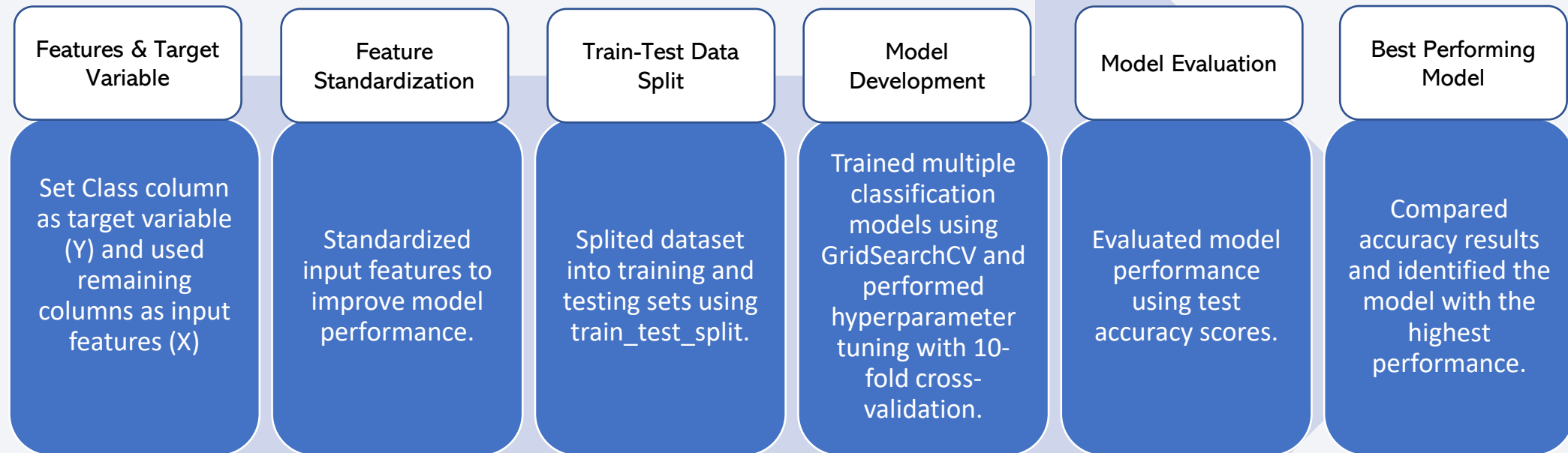
- Added a range slider to allow users to filter launches based on payload mass.
- This enables analysis of how payload size influences mission outcomes.

- **Success Payload Scatter Plot Callback**

- Added a callback function to generate a scatter plot showing the relationship between payload mass and launch success.
- Booster versions were colour-coded to identify performance differences between booster types.

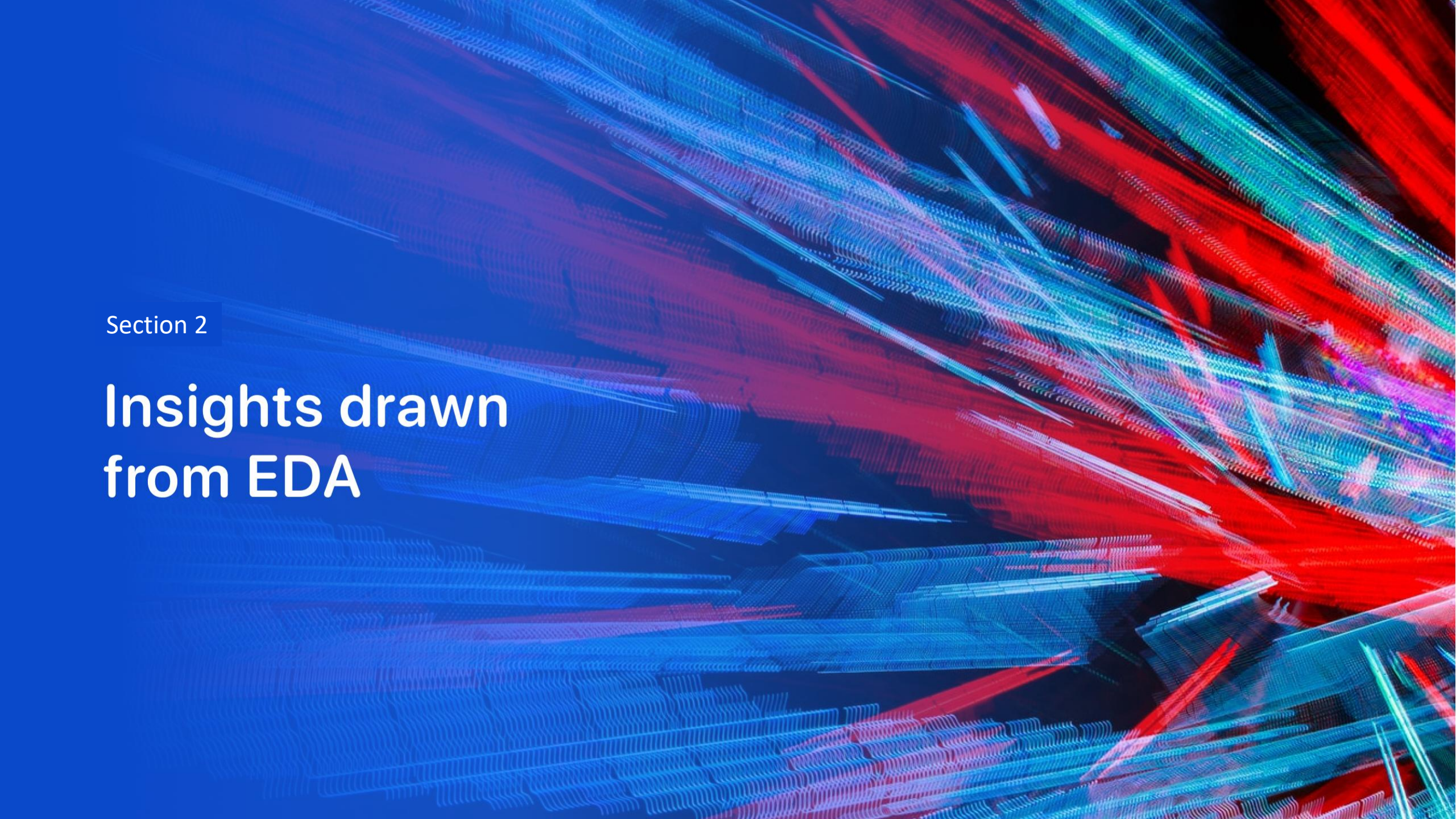
[Notebook GitHub URL](#)

Predictive Analysis (Classification)



Results

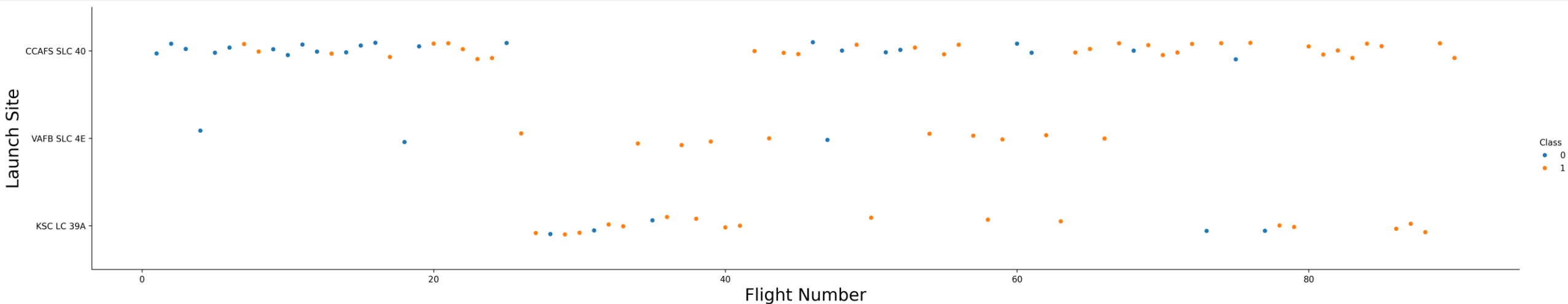
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

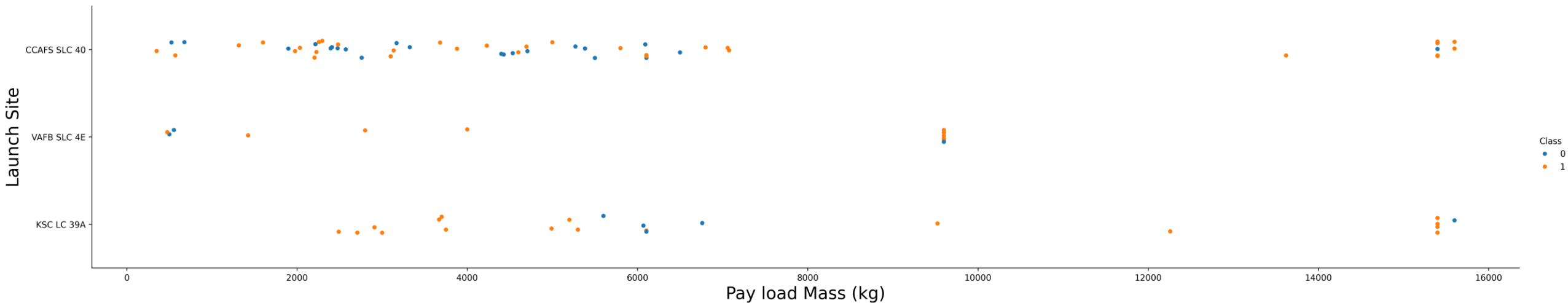
Insights drawn from EDA

Flight Number vs. Launch Site



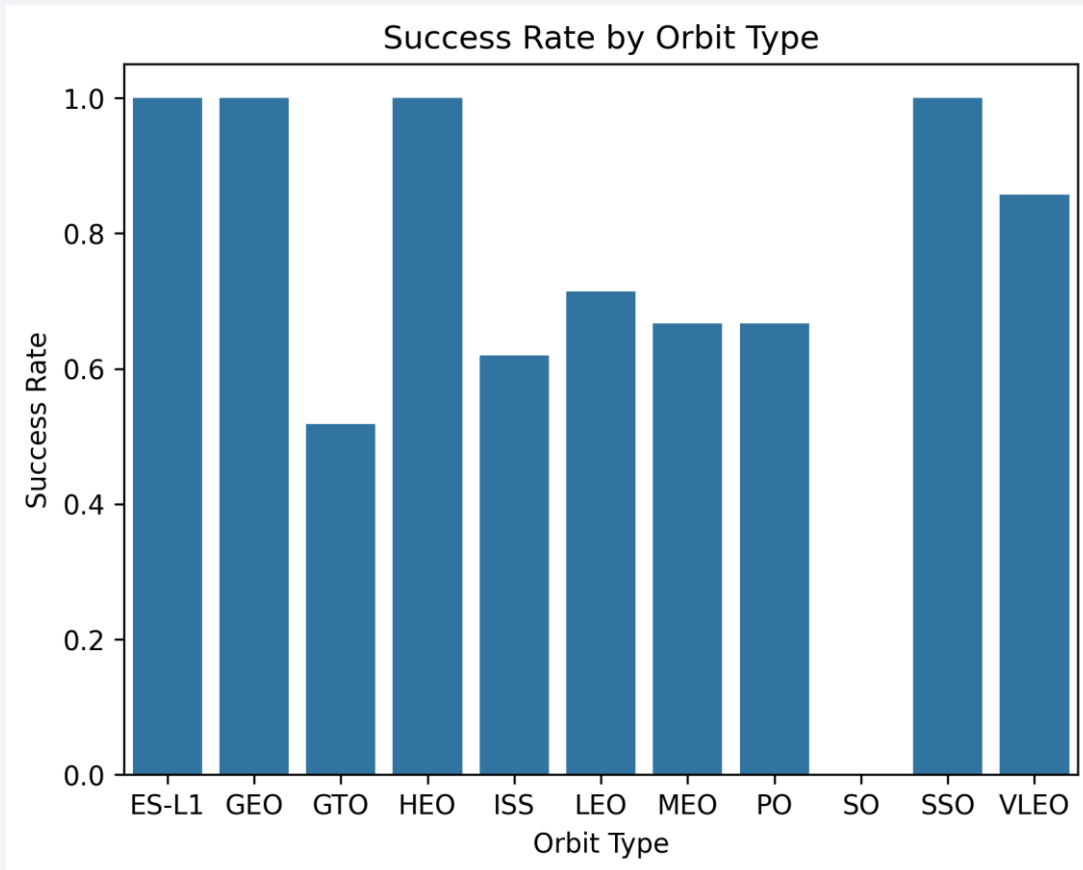
Most launches have been conducted from the CCAFS SLC 40 launch site. The plot shows that as the number of flights increased over time, the proportion of successful launches generally improved across all launch sites, indicating improved reliability of Falcon 9 first-stage landings.

Payload vs. Launch Site



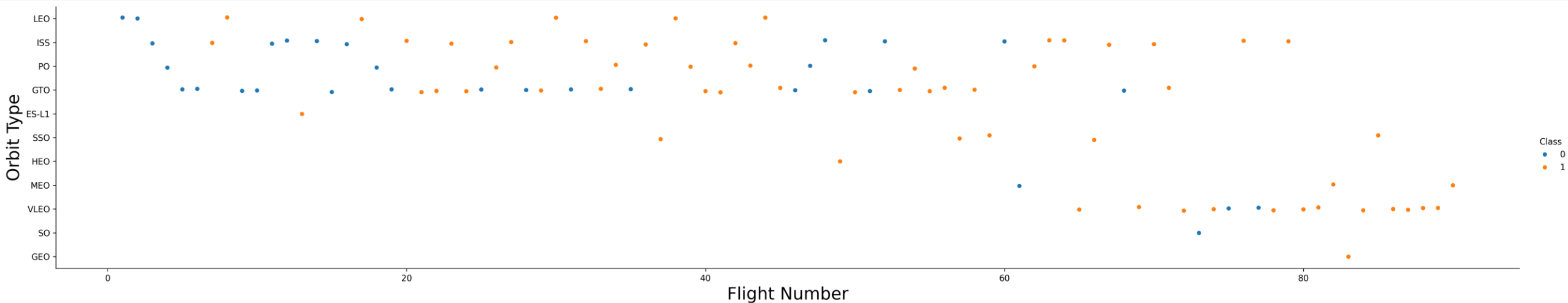
For the VAFB-SLC launch site there are no rockets launched for heavy payload mass (greater than 10000).

Success Rate vs. Orbit Type



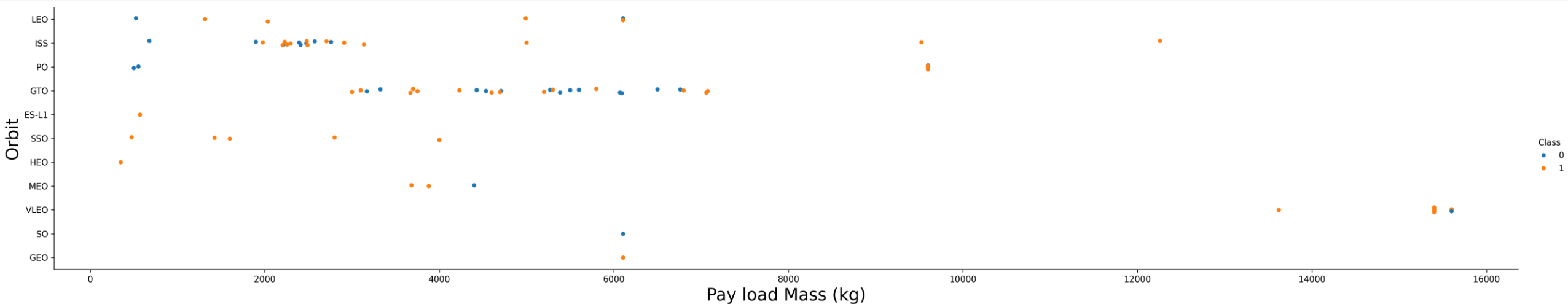
Orbits like ES-L1, GEO, HEO, and SSO have a perfect 100% success rate, while VLEO is also very high at over 80%. On the other hand, GTO has the lowest success rate at around 50%, and SO has a 0% success rate, showing that the target orbit heavily impacts whether a launch succeeds.

Flight Number vs. Orbit Type



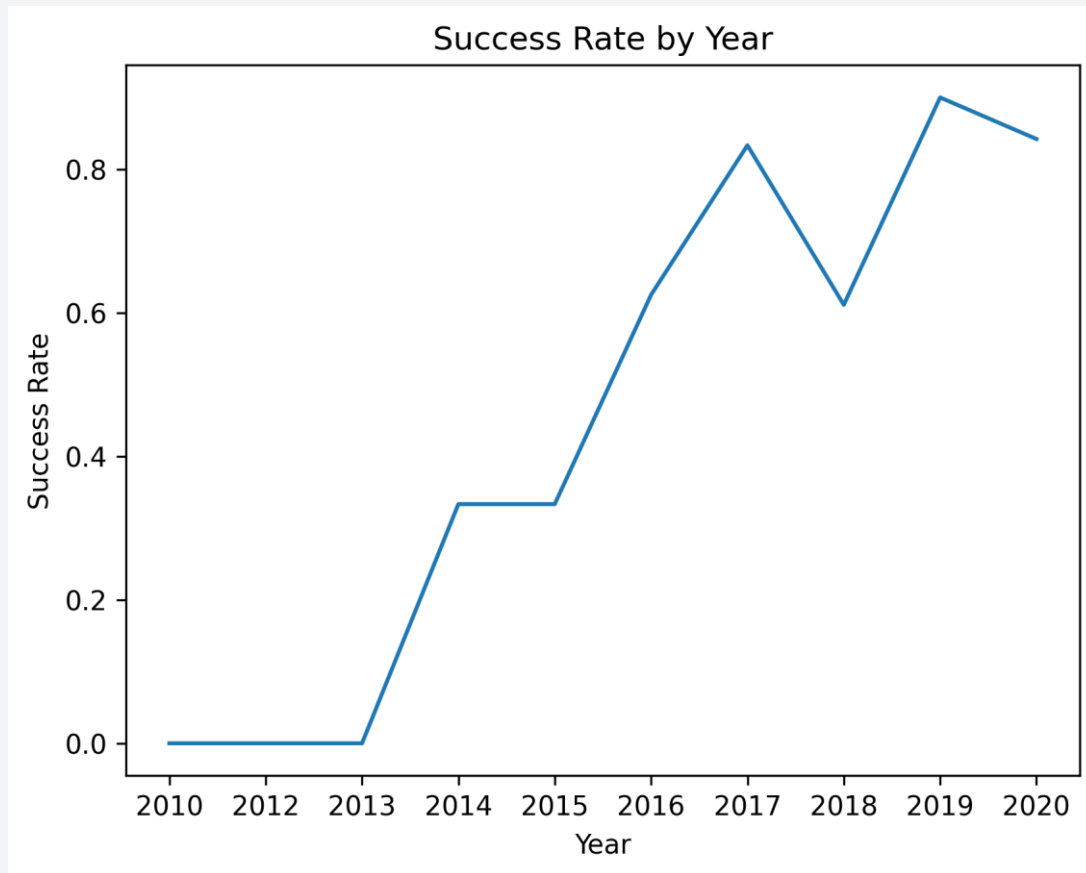
In the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.

Payload vs. Orbit Type



With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS. However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.

Launch Success Yearly Trend



The launch success rate showed a steady upward trend from 2013 to 2020. Although there was a temporary drop in 2018 down to about 60%, the rate quickly recovered in 2019 to peak at over 90%.

All Launch Site Names

Query	<pre>%%sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;</pre>						
Description	This SQL query finds a list of all the different launch locations used in the dataset. Because it uses the word `DISTINCT`, it removes all duplicates. This means if a launch site was used many times, it will only show up once in the final list. It looks inside the `Launch_Site` column and gets this information from the table called `SPACEXTABLE`. In short, it gives you a clean list of the unique launch pads without repeating any names.						
Results	<table><tr><th>Launch_Site</th></tr><tr><td>CCAFS LC-40</td></tr><tr><td>VAFB SLC-4E</td></tr><tr><td>KSC LC-39A</td></tr><tr><td>CCAFS SLC-40</td></tr></table>	Launch_Site	CCAFS LC-40	VAFB SLC-4E	KSC LC-39A	CCAFS SLC-40	
Launch_Site							
CCAFS LC-40							
VAFB SLC-4E							
KSC LC-39A							
CCAFS SLC-40							

Launch Site Names Begin with 'CCA'

Query	<pre>%%sql SELECT * FROM SPACEXTABLE WHERE "Launch_site" LIKE 'CCA%' LIMIT 5</pre>									
Description	This SQL query retrieves all the available information for the first five records in the dataset where the launch location's name begins with the letters "CCA". By using `SELECT *`, the query pulls every column of data from `SPACEXTABLE`, while the `LIKE 'CCA%'` condition filters the results to only include sites that start with those specific letters. Finally, the `LIMIT 5` clause ensures that the output stays short and manageable by stopping the search as soon as the first five matching rows are found.									
Results	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG	Orbit	Customer	Mission_Outcome	Landing_Outcome
	2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
	2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
	2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
	2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
	2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

Query	<pre>%%sql SELECT SUM("PAYLOAD_MASS__KG_") AS total_payload_mass FROM SPACEXTABLE WHERE "Customer" = 'NASA (CRS)';</pre>			
Description	This SQL query calculates the total sum of the ``PAYLOAD_MASS__KG_`` column from the table named `SPACEXTABLE`. It uses a `WHERE` clause to only include rows where the ``"Customer"`` column is exactly equal to `NASA (CRS)`, and renames the resulting calculated number as `total_payload_mass`.			
Results	<table><tr><th>total_payload_mass</th></tr><tr><td>45596</td></tr></table>	total_payload_mass	45596	
total_payload_mass				
45596				

Average Payload Mass by F9 v1.1

Query	<pre>%%sql SELECT AVG("PAYLOAD_MASS__KG_") AS average_payload_mass FROM SPACEXTABLE WHERE "BOOSTER_VERSION" = 'F9 v1.1';</pre>	
Description	This SQL query calculates the average value of the <code>"PAYLOAD_MASS__KG_"</code> column from the table named <code>SPACEXTABLE</code> . It uses a <code>WHERE</code> clause to only include rows where the <code>"BOOSTER_VERSION"</code> column is exactly equal to <code>'F9 v1.1'</code> , and renames the resulting calculated number as <code>average_payload_mass</code> .	
Results	average_payload_mass 2928.4	

First Successful Ground Landing Date

Query	<pre>%%sql SELECT MIN("Date") AS first_successful_ground_landing FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';</pre>	
Description	This SQL query finds the minimum (earliest) value in the `"Date"` column from the table named `SPACEXTABLE`. It uses a `WHERE` clause to only include rows where the `"Landing_Outcome"` column is exactly equal to `Success (ground pad)`, and renames the resulting calculated value as `first_successful_ground_landing`.	
Results	first_successful_ground_landing	
	2015-12-22	

Successful Drone Ship Landing with Payload between 4000 and 6000

Query	<pre>%%sql SELECT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)' AND "PAYLOAD_MASS_KG_" > 4000 AND "PAYLOAD_MASS_KG_" < 6000;</pre>						
Description	This SQL query selects values from the `Booster_Version` column in the table named `SPACEXTABLE`. It uses a `WHERE` clause to filter for rows where the `Landing_Outcome` column is exactly equal to `Success (drone ship)`, and the `PAYLOAD_MASS_KG_` column is both greater than `4000` and less than `6000`.						
Results	<table><tr><th>Booster_Version</th></tr><tr><td>F9 FT B1022</td></tr><tr><td>F9 FT B1026</td></tr><tr><td>F9 FT B1021.2</td></tr><tr><td>F9 FT B1031.2</td></tr></table>	Booster_Version	F9 FT B1022	F9 FT B1026	F9 FT B1021.2	F9 FT B1031.2	
Booster_Version							
F9 FT B1022							
F9 FT B1026							
F9 FT B1021.2							
F9 FT B1031.2							

Total Number of Successful and Failure Mission Outcomes

Query	<pre>%%sql SELECT TRIM("Mission_Outcome") AS Clean_Outcome, COUNT(*) AS Total_Count FROM SPACEXTABLE GROUP BY Clean_Outcome;</pre>									
Description	This SQL query selects the trimmed values of the ``Mission_Outcome`` column, renaming it as `clean_Outcome`, alongside the total count of rows for each outcome from the table named `SPACEXTABLE`. It uses a `GROUP BY` clause to group the records by the `clean_Outcome` value so it can provide a count for each distinct category.									
Results	<table><tr><th>Clean_Outcome</th><th>Total_Count</th></tr><tr><td>Failure (in flight)</td><td>1</td></tr><tr><td>Success</td><td>99</td></tr><tr><td>Success (payload status unclear)</td><td>1</td></tr></table>	Clean_Outcome	Total_Count	Failure (in flight)	1	Success	99	Success (payload status unclear)	1	
Clean_Outcome	Total_Count									
Failure (in flight)	1									
Success	99									
Success (payload status unclear)	1									

Boosters Carried Maximum Payload

Query	<pre>%%sql SELECT "BOOSTER_VERSION", "PAYLOAD_MASS_KG_" FROM SPACEXTABLE WHERE "PAYLOAD_MASS_KG_" = (SELECT MAX("PAYLOAD_MASS_KG_") FROM SPACEXTABLE);</pre>																											
Description	This SQL query selects values from the ``BOOSTER_VERSION`` and ``PAYLOAD_MASS_KG_`` columns in the table named `SPACEXTABLE`. It uses a `WHERE` clause to only include rows where the ``PAYLOAD_MASS_KG_`` column is exactly equal to the maximum value found in that same column across the entire table.																											
Results	<table><tr><th>Booster_Version</th><th>PAYLOAD_MASS_KG_</th></tr><tr><td>F9 B5 B1048.4</td><td>15600</td></tr><tr><td>F9 B5 B1049.4</td><td>15600</td></tr><tr><td>F9 B5 B1051.3</td><td>15600</td></tr><tr><td>F9 B5 B1056.4</td><td>15600</td></tr><tr><td>F9 B5 B1048.5</td><td>15600</td></tr><tr><td>F9 B5 B1051.4</td><td>15600</td></tr><tr><td>F9 B5 B1049.5</td><td>15600</td></tr><tr><td>F9 B5 B1060.2</td><td>15600</td></tr><tr><td>F9 B5 B1058.3</td><td>15600</td></tr><tr><td>F9 B5 B1051.6</td><td>15600</td></tr><tr><td>F9 B5 B1060.3</td><td>15600</td></tr><tr><td>F9 B5 B1049.7</td><td>15600</td></tr></table>	Booster_Version	PAYLOAD_MASS_KG_	F9 B5 B1048.4	15600	F9 B5 B1049.4	15600	F9 B5 B1051.3	15600	F9 B5 B1056.4	15600	F9 B5 B1048.5	15600	F9 B5 B1051.4	15600	F9 B5 B1049.5	15600	F9 B5 B1060.2	15600	F9 B5 B1058.3	15600	F9 B5 B1051.6	15600	F9 B5 B1060.3	15600	F9 B5 B1049.7	15600	
Booster_Version	PAYLOAD_MASS_KG_																											
F9 B5 B1048.4	15600																											
F9 B5 B1049.4	15600																											
F9 B5 B1051.3	15600																											
F9 B5 B1056.4	15600																											
F9 B5 B1048.5	15600																											
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F9 B5 B1060.2	15600																											
F9 B5 B1058.3	15600																											
F9 B5 B1051.6	15600																											
F9 B5 B1060.3	15600																											
F9 B5 B1049.7	15600																											

2015 Launch Records

Query	<pre>%%sql SELECT CASE substr("Date", 6, 2) WHEN '01' THEN 'January' WHEN '02' THEN 'February' WHEN '03' THEN 'March' WHEN '04' THEN 'April' WHEN '05' THEN 'May' WHEN '06' THEN 'June' WHEN '07' THEN 'July' WHEN '08' THEN 'August' WHEN '09' THEN 'September' WHEN '10' THEN 'October' WHEN '11' THEN 'November' WHEN '12' THEN 'December' END AS Month_Name, "Landing_Outcome", "Booster_Version", "Launch_Site" FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Failure (drone ship)' AND substr("Date", 1, 4) = '2015';</pre>												
Description	This SQL query selects a text name for the month by extracting a two-character substring from the "Date" column, alongside the "Landing_Outcome", "Booster_Version", and "Launch_Site" columns from the table named SPACEXTABLE. It uses a WHERE clause to only include rows where the "Landing_Outcome" column is exactly equal to 'Failure (drone ship)' and the first four characters of the "Date" column are exactly equal to '2015'.												
Results	<table><tr><th>Month_Name</th><th>Landing_Outcome</th><th>Booster_Version</th><th>Launch_Site</th></tr><tr><td>January</td><td>Failure (drone ship)</td><td>F9 v1.1 B1012</td><td>CCAFS LC-40</td></tr><tr><td>April</td><td>Failure (drone ship)</td><td>F9 v1.1 B1015</td><td>CCAFS LC-40</td></tr></table>	Month_Name	Landing_Outcome	Booster_Version	Launch_Site	January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40	April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40
Month_Name	Landing_Outcome	Booster_Version	Launch_Site										
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40										
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40										

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Query	<pre>%%sql SELECT "LANDING_OUTCOME", COUNT(*) AS OUTCOME_COUNT FROM SPACEXTABLE WHERE "DATE" BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY "LANDING_OUTCOME" ORDER BY OUTCOME_COUNT DESC;</pre>																			
Description	This SQL query selects the "LANDING_OUTCOME" column alongside the total count of rows for each outcome, renamed as OUTCOME_COUNT, from the table named SPACEXTABLE. It uses a WHERE clause to only include rows where the "DATE" column falls between '2010-06-04' and '2017-03-20'. Finally, it uses a GROUP BY clause to group the records by the "LANDING_OUTCOME" value and uses an ORDER BY clause to sort the results by OUTCOME_COUNT in descending order.																			
Results	<table><tr><th>Landing_Outcome</th><th>OUTCOME_COUNT</th></tr><tr><td>No attempt</td><td>10</td></tr><tr><td>Success (drone ship)</td><td>5</td></tr><tr><td>Failure (drone ship)</td><td>5</td></tr><tr><td>Success (ground pad)</td><td>3</td></tr><tr><td>Controlled (ocean)</td><td>3</td></tr><tr><td>Uncontrolled (ocean)</td><td>2</td></tr><tr><td>Failure (parachute)</td><td>2</td></tr><tr><td>Precluded (drone ship)</td><td>1</td></tr></table>	Landing_Outcome	OUTCOME_COUNT	No attempt	10	Success (drone ship)	5	Failure (drone ship)	5	Success (ground pad)	3	Controlled (ocean)	3	Uncontrolled (ocean)	2	Failure (parachute)	2	Precluded (drone ship)	1	
Landing_Outcome	OUTCOME_COUNT																			
No attempt	10																			
Success (drone ship)	5																			
Failure (drone ship)	5																			
Success (ground pad)	3																			
Controlled (ocean)	3																			
Uncontrolled (ocean)	2																			
Failure (parachute)	2																			
Precluded (drone ship)	1																			

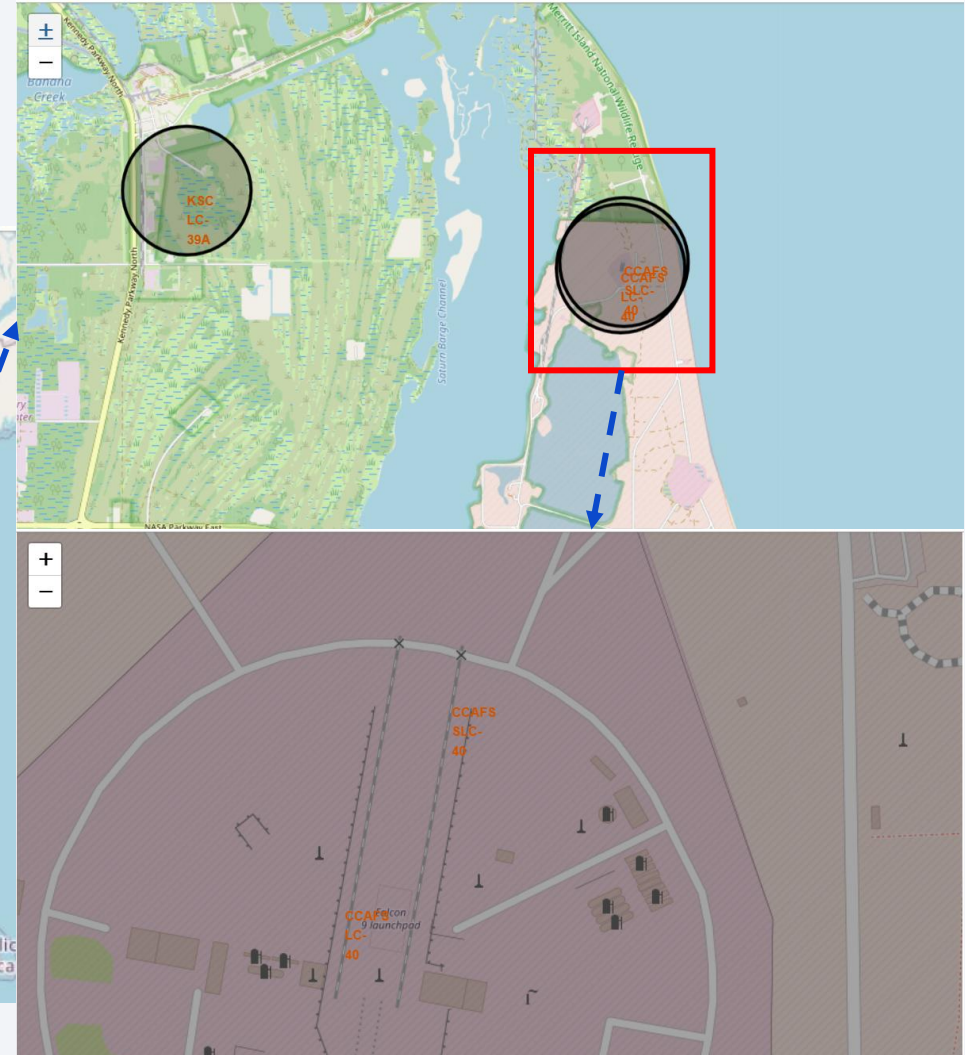
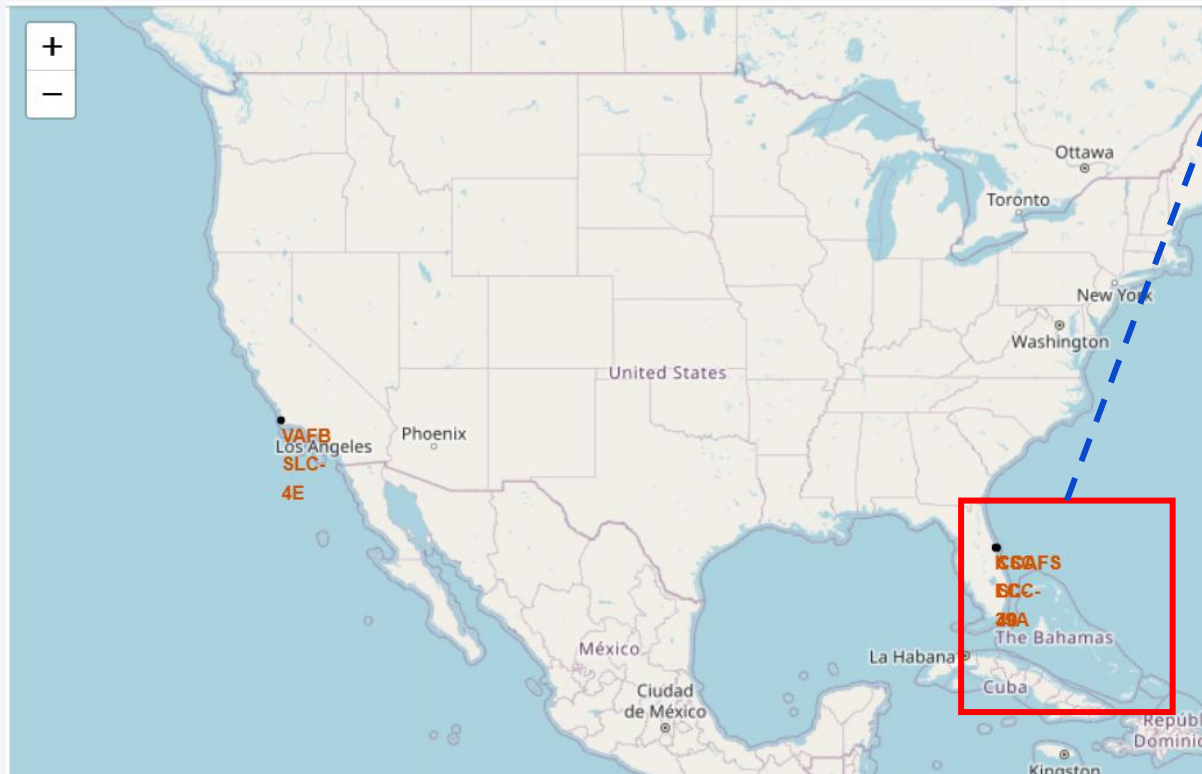
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

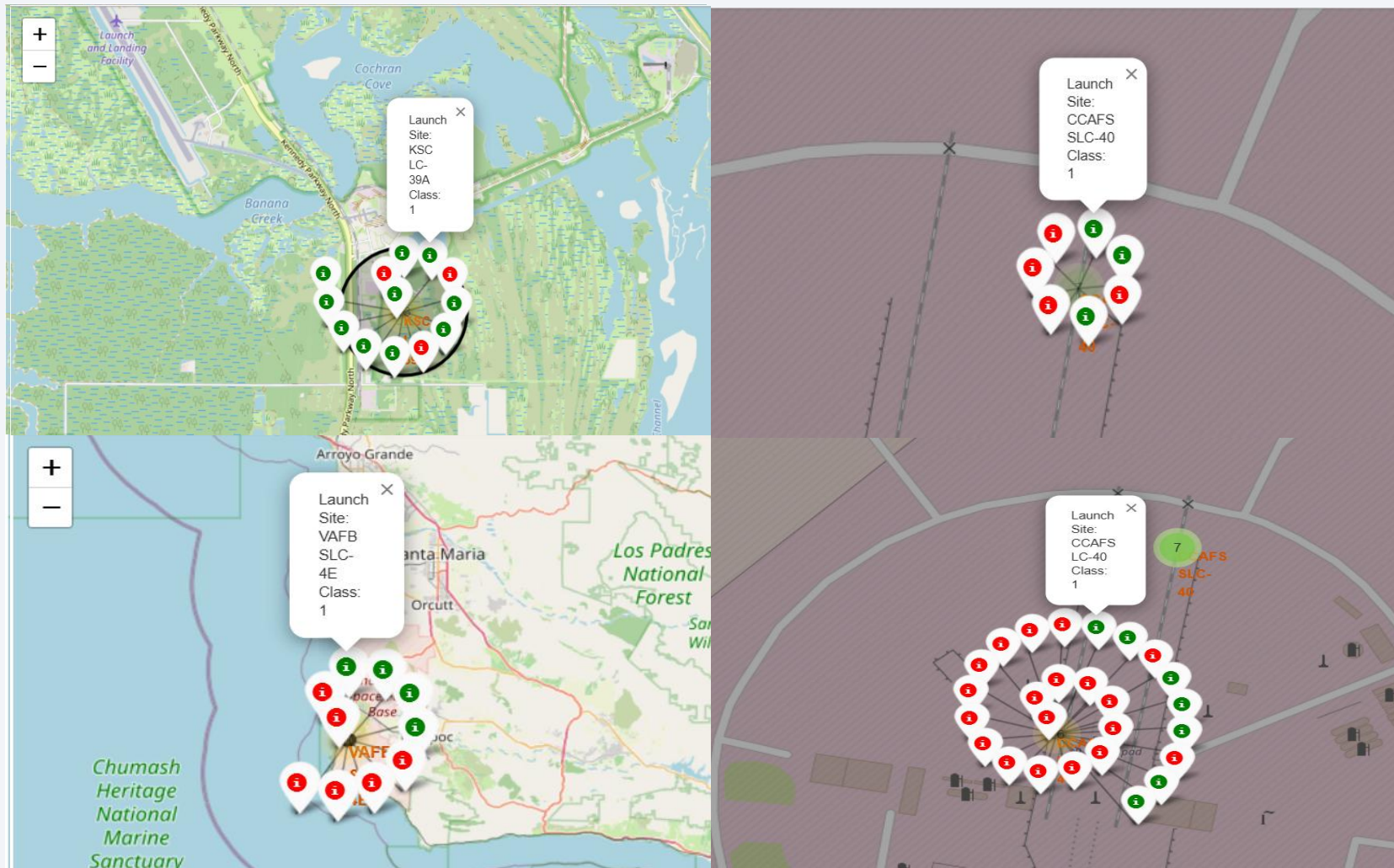
Launch Sites

It is interesting to note all launch sites are near the water and equator line.



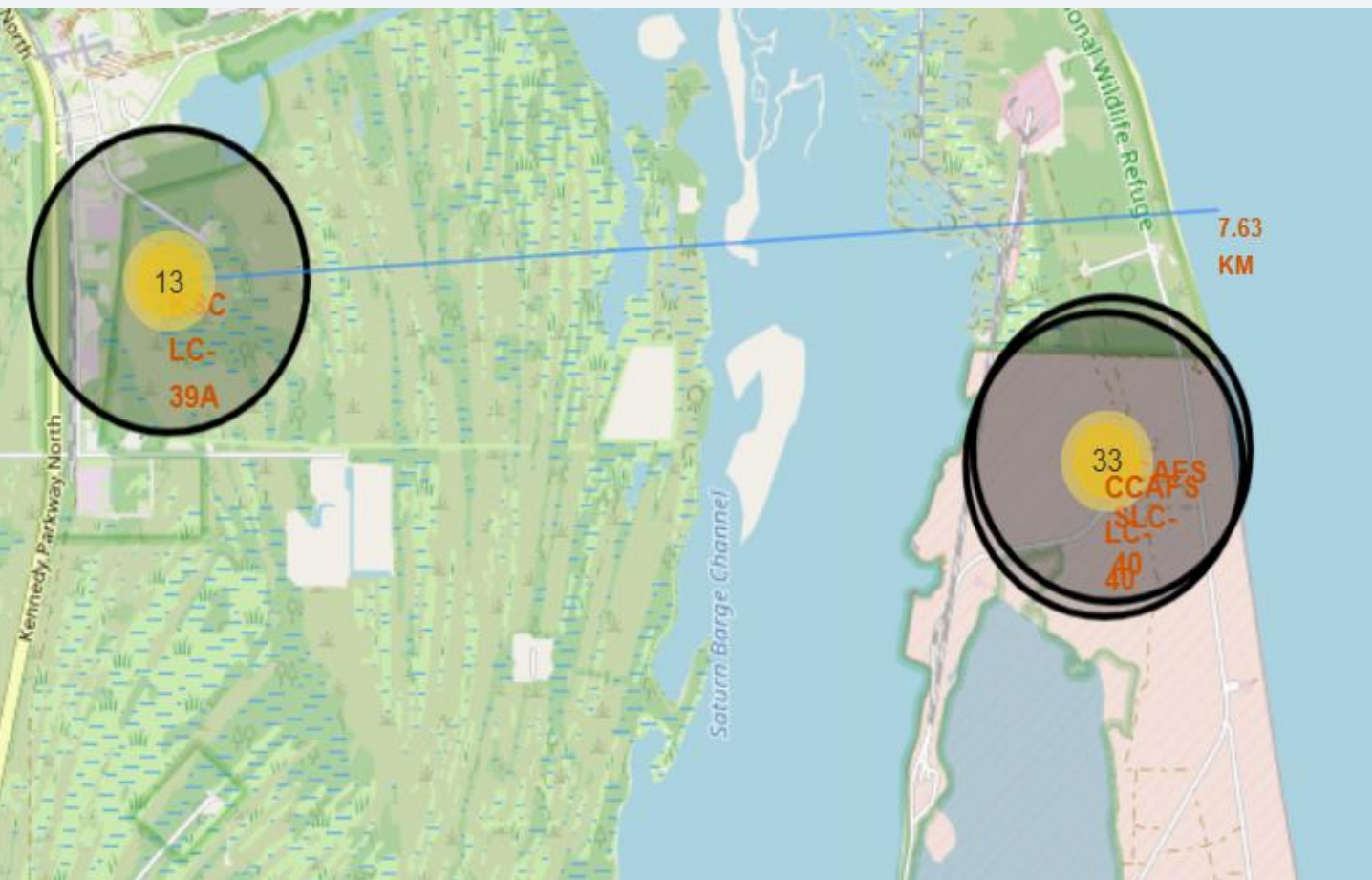
Launch Success/Failure by Site

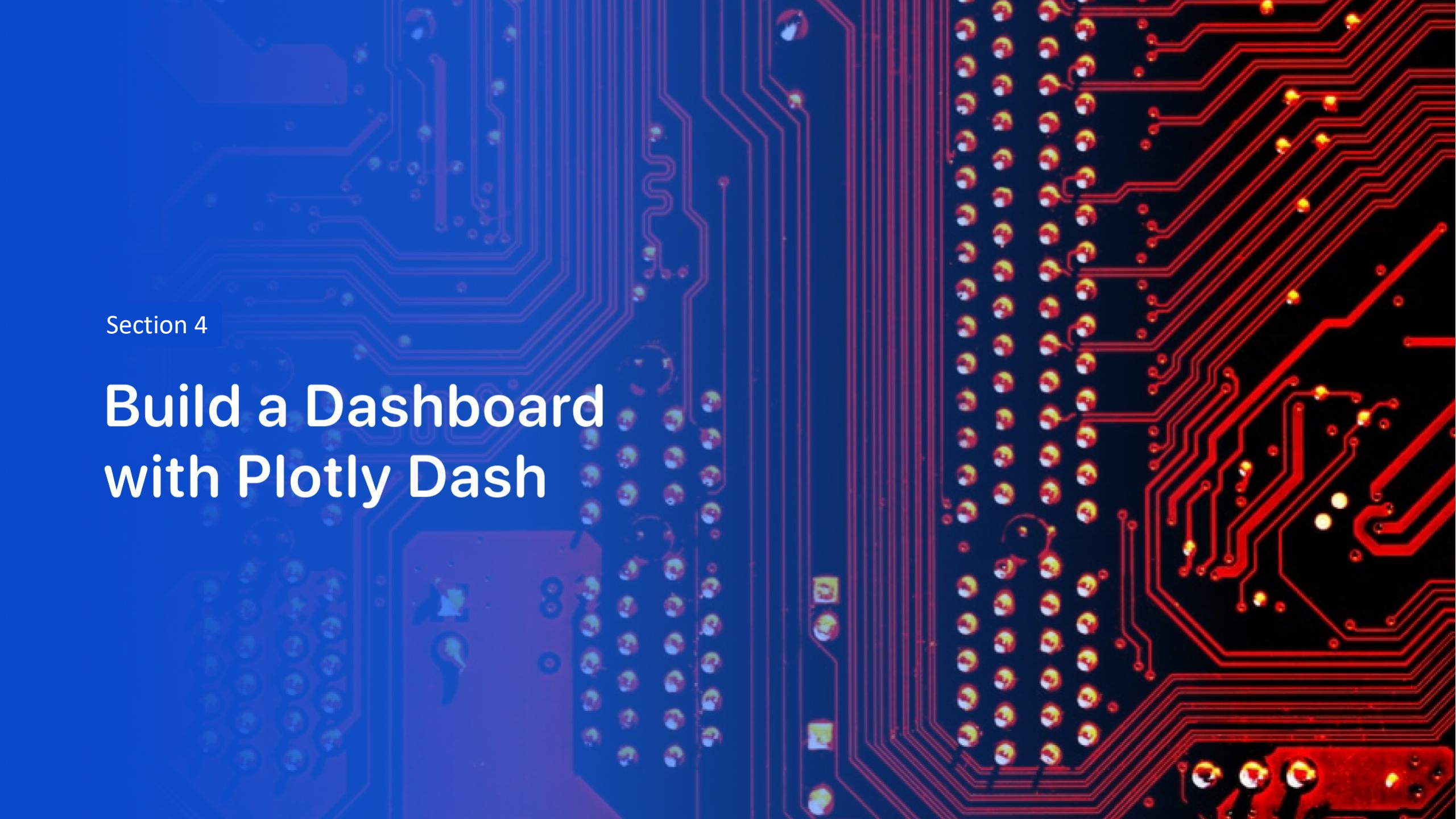
These maps show where the most rockets are launched and highlight the high success rate at KSC LC-39A.



Proximity Analysis of Launch Sites

All launch sites are in close approximately of coast, highways, and railroads and far away from big cities.



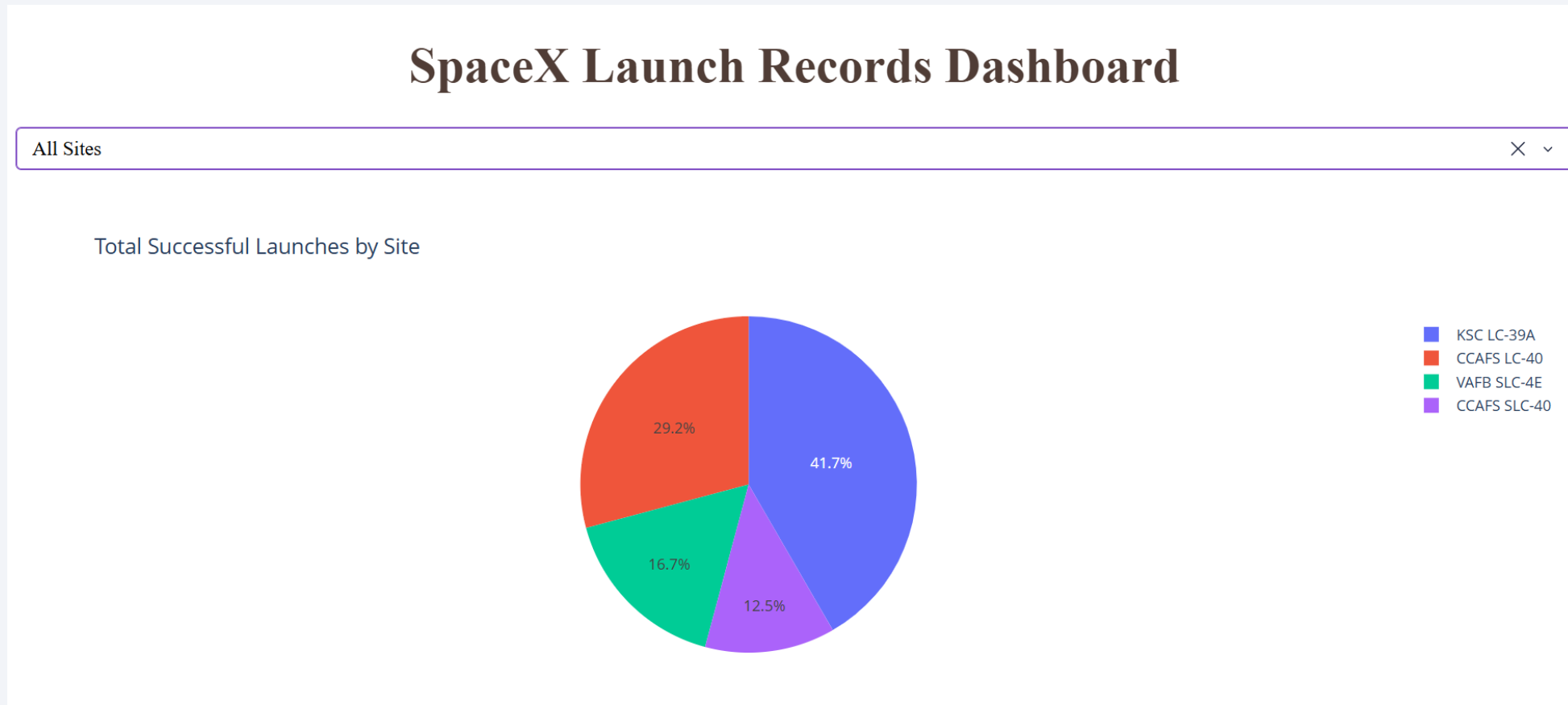


Section 4

Build a Dashboard with Plotly Dash

Successful Falcon 9 First-Stage Landings by Launch Site

KSC LC-39A recorded the highest proportion of successful Falcon 9 first-stage landings (41.7%), followed by CCAFS LC-40 (29.2%). VAFB SLC-4E and CCAFS SLC-40 accounted for 16.7% and 12.5% of successful first-stage landings, respectively.



Falcon 9 First-Stage Landing Success at KSC LC-39A

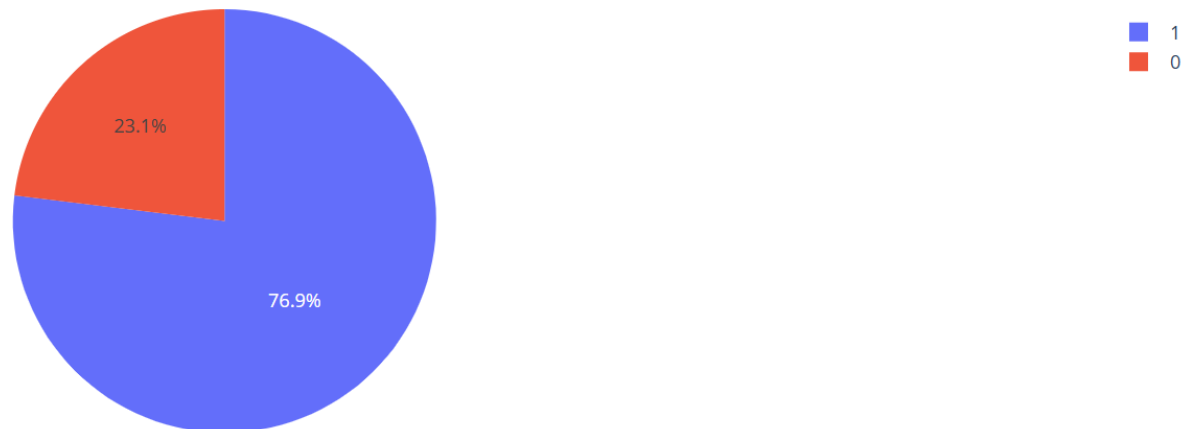
Approximately 76.9% of Falcon 9 missions at KSC LC-39A achieved a successful first-stage landing, while 23.1% resulted in a failed landing.

SpaceX Launch Records Dashboard

KSC LC-39A

× ▾

Success vs. Failed Launches for KSC LC-39A



Relationship Between Payload Mass and Falcon 9 First-Stage Landing Success

Most launches occurred between 2,000 and 6,000 kg of payload mass, with both successful and failed landings. Newer booster versions, particularly FT, B4, and B5, achieved a higher proportion of successful first-stage landings, indicating improved landing reliability over time.

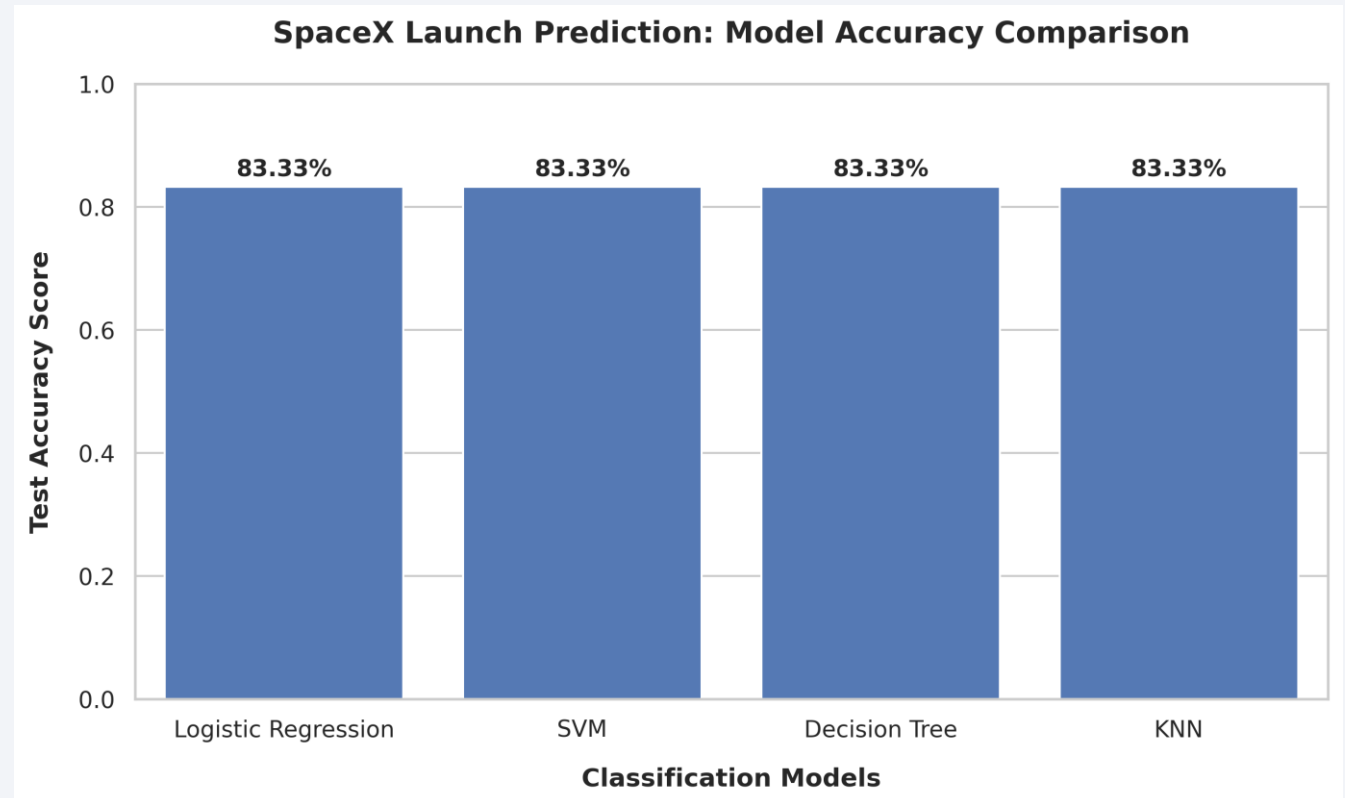


Section 5

Predictive Analysis (Classification)

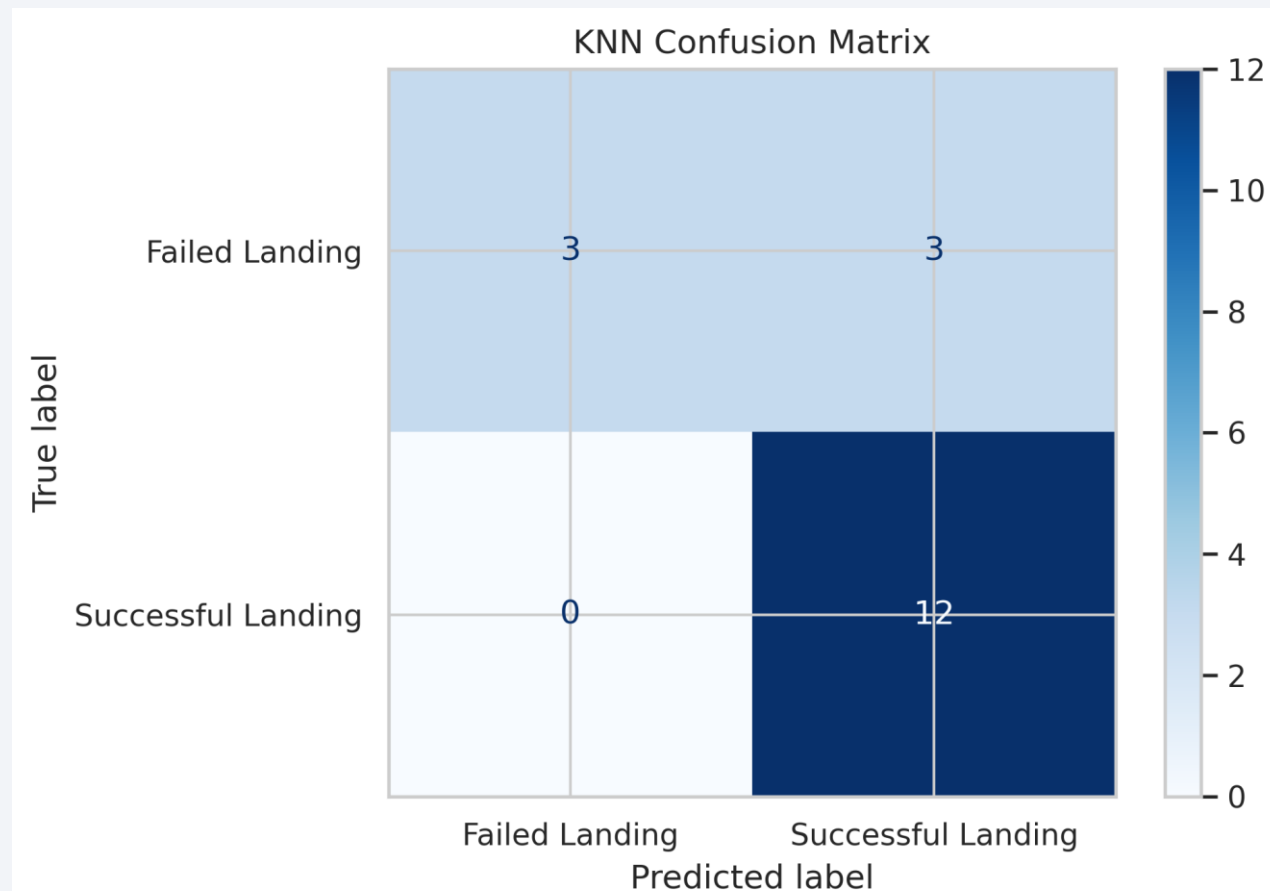
Classification Accuracy

All four classification models, Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN), achieved the highest and identical test accuracy of 83.33%. Therefore, based solely on classification accuracy, no single model demonstrated superior performance. Further evaluation using additional performance metrics, such as precision, recall, F1-score, and confusion matrix analysis, is required to identify the most suitable model for predicting successful Falcon 9 first-stage landings.



Confusion Matrix

The confusion matrices show that all models correctly predicted 12 successful landings and 3 failed landings, with 3 false positives and no false negatives.



Conclusions

- Predictive classification machine learning models can successfully predict Falcon 9 first-stage landing outcomes. However, prediction accuracy depends on the size of the dataset and how well the model is developed.
- Launch site, orbit type, and payload mass all have an important effect on first-stage landing success. These factors are also related to each other and can influence one another.
- The first-stage landing success rate has improved over time, with an increasing number of flights and newer Falcon 9 booster versions achieving higher success rates.
- These results show that historical launch data can be used to predict landing success, helping improve decisions about first-stage reusability and reducing launch costs.

Appendix

- <https://api.spacexdata.com/v4/launches/past>
- <https://api.spacexdata.com/v4/rockets/>
- <https://api.spacexdata.com/v4/launchpads/>
- <https://api.spacexdata.com/v4/payloads/>
- <https://api.spacexdata.com/v4/cores/>
- https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches

Thank you!

